MIL-A-70608A(AR)
31 May 1988
SUPERSEDING
MIL-A-70608(AR)
6 June 1986

MILITARY SPECIFICATION

ADAPTER & FUZE, ASSEMBLY: M624, LOADING, ASSEMBLING AND PACKING

This Specification is approved for use within the U. S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers quality assurance provisions and preparation for delivery for the loading and assembling of Adapter & Fuze, Assembly: M624.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, listed in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-2550 -	General Specification for Ammunition Except Small Arms Ammunition.
MIL-A-48078-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-D-70603-	Detonator, Stab, PA523, Loading, Assembling, and Packing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament,Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 1345
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STANDARDS

MILITARY

MIL-STD-105- Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-331- Fuze and Fuze Components, Environmental and Performance Tests For

MIL-STD-1235- Continuous Sampling Procedures and Table for Inspection by Attributes.

(Copies of specifications, standards, drawings, publications, and other Government documents required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12546032- Adapter & Fuze, Assembly- M624

12556001- Packing and Marking For Box Ammunition M19A1 For Adapter and Fuze, Assembly, M624.

12556005- Packing and Marking For Box, Wirebound For Fuze Adapter and Fuze Assembly, M624.

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings, specifications and standards.

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3.2 Assemblies. The assemblies shall comply with all requirements specified on Drawing (dwg.) 12546032, and with all requirements specified in applicable specifications.

3.3 Jolt. The assemblies shall withstand the jolt test specified in MIL-STD-331 when tested as specified in 4.5.1.

3.4 Jumble. The assemblies shall withstand the jumble test specified in MIL-STD-331 when tested as specified in 4.5.2.

3.5 X-ray of assembly. All assemblies without extension rod assembly extension shall be x-rayed in accordance with 4.5.3.

3.6 Energy output of assembly. Assemblies shall be tested for energy output in accordance with 4.5.4.

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of cracks, gouges, burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly, nor shall they be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection.

b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated below and Table I.

Part Description	Drawings	Quantities
Adapter & Fuze assembly	12546032	62

The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3. Unless otherwise directed by the Contracting Officer, loaded components shall not be submitted to a Proving Ground for any test until the above assemblies submitted in accordance with this paragraph have been accepted.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein. If any items fail to comply with any of the applicable requirements, the first article sample shall be rejected.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-A-70608A(AR)

PARAGRAPH	TITLE: Adapter & Fuze, Assembly: M624		SHEET 1 OF 1		DRAWING NUMBER: See below
					NEXT HIGHER ASSEMBLY: --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Adapter & fuze assembly: M624 (Dwg. 12546032)				
	Examination for defects	62		3.2	4.4.2.1
	Xray	S0-0-1	(1)	3.5	4.5.3
	Energy output tests	S0-0-1	(1)	3.6	4.5.4
	Jolt	6-0-1		3.3	4.5.1
	Jumble	6-0-1		3.4	4.5.2

NOTES:
1. X-ray prior to energy output test.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.3.3 Rejection. See MIL-A-48078. If any items fail to comply with any of the applicable requirements, the first article sample shall be rejected.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.1.1 Cavity identification (plastic and cast parts). The contractor shall maintain an approved cavity identification system. Components shall be grouped and identified until all inspections have been completed.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Tests Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70608A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.1	Adapter & Fuze, Assembly: M624		12546032
			NEXT HIGHER ASSEMBLY 12546034

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Boxed end pin missing		100%	3.2	Visual
2	Xray		100%(2)	3.5	4.5.3
3	Jolt	6-0-1	(1)	3.3	4.5.1
4	Jumble	6-0-1	(1)	3.4	4.5.2
5	Cap assembly missing or improperly secured		100%	3.2	Visual/Manual
6	Band and stop missing		100%	3.2	Visual
Major					
101	Components missing (not including delay assembly of compression spring.)		0.40%	3.2	Visual
102	Assembly damaged (dented, deformed, cracked, or punctured		0.40%	3.2	Visual
103	Jolt	6-0-1	(1)	3.3	4.5.1
104	Jumble	6-0-1	(1)	3.4	4.5.2
105	Energy output test		0.40%	3.6	4.5.4
106	X-ray		100%(1)	3.5	4.5.3

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.1	Adapter & Fuze, Assembly: M624				12546032
					NEXT HIGHER ASSEMBLY 12546034
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Loose joint between fuze and adapter		0.65%	3.2	Manual
202	Protective coating damaged (other than light scratches)		0.65%	3.2	Visual
203	Marking misleading or unidentifiable		0.65%	3.2	Visual
204	Poor workmanship		1.0%	3.8	Visual

NOTES: (1) Samples which have detonator firing during tests shall be classed as critical defects. Samples which fail for cause other than detonator firing shall be classed as major defects. (2) Critical defects - proper seating of tilt rod, presence of tilt rod groove pin, presence of firing pin spring assembly groove pin and proper position of firing pin spring assembly. Major Defects - presence of the delay assembly and compression spring.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70608A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Box, Ammunition: M19A1 prior to sealing	1 OF 1	12556001
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Contents incorrect or missing		1.5%	3.1	Visual
202	Fillers missing		1.5%	3.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70608A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Box, Ammunition: M19A1 after sealing	1 OF 1	12556001
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged		1.0%	3.1	Visual
102	Contents of box exposed		1.0%	3.1	Visual
Minor					
201	Contents loose		1.5%	3.1	Manual/Visual
202	Box inadequately or improperly sealed		1.5%	3.1	Visual
203	Marking missing, incorrect or illegible		1.5%	3.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70608A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Box, Wirebound				12556005
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Hardware missing, broken or loose		0.40%	3.1	Visual/Manual
102	Fillers missing		0.40%	3.1	Visual
103	Box damaged so that contents are, exposed or liable to become exposed		0.40%	3.1	Visual
104	DOD symbol misleading or unidentifiable		0.40%	3.1	Visual
Minor					
201	Hardware improperly engaged		0.65%	3.1	Visual/Manual
202	Car seal missing, unsealed or improperly positioned		0.65%	3.1	Visual/Manual
203	Contents loose when box is tilted		0.65%	3.1	Visual/Manual
204	Marking misleading or unidentifiable		0.65%	3.1	Visual

NOTES:

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4.4.3 Testing.

4.4.3.1 Check test for deterioration of detonators (see Specification MIL-D-70603. If the total time between original acceptance of any detonator lot and assembly of that lot into the mine fuze exceeds two years, or if the detonators have been subjected to adverse conditions, however brief, at any time since previous tests, the detonator lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-D-70603 immediately before the lot is assembled into the fuzes. This test shall be performed by the contractor on detonators selected by the Government inspector at the facility assembling the detonators into the fuzes. The test shall be conducted at Government expense, without cost to the contractor who loaded the detonator, or to the contractor assembling the detonators into the fuzes, and shall not constitute a basis of rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the lot was under jurisdiction of either contractor.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed in this specification is identified in the "Paragraph Reference/Inspection Method" column in Classification of Defects and Tests forms beginning with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Jolt. The adapter and fuze assembly (minus the extension rod assembly) must be assembled to a specified fixture and subjected to the jolt test as specified in MIL-STD-331. Upon completion of the test, the fuze shall be examined for detonator firing, unseating of the detonator or other listed defects by removing the boxed end pin, band and stop and the cap assembly.

4.5.2 Jumble. Teh adapter and fuze assembly (minus the extension rod assembly) shall be subjected to the jumble test as specified in MIL-STD-331. Upon completion of the test, the fuze shall be examined for detonator firing, unseating of the detonator or other listed defects by removing the boxed end pin, band and stop and the cap assembly.

4.5.3 X-ray of assembly. The entire production of loaded assembies, less extension rod assemblies, shall be x-rayed to confirm proper seating of the tilt rod on the firing pin spring assembly, presence of the tilt rod groove pin and firing pin spring assembly groove pin, and to assure firing pin spring assemby is in the correct position for firing. In addition, confirm presence of the delay assembly and compression spring.

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4.5.4 Energy output test. Loaded assemblies shall be placed in a test fixture. Assemble the Extension Rod Assembly into the pressure ring. The cap assembly, boxed end pin and band and stop shall be removed. A uniform load shall be applied at the end of the extension rod assembly causing detonation against a witness plate (1 1/4" x 1 1/4" x 5/8" steel). The dent shall be measured after detonations. The minimum dent shall be 0.020 inch.

5. PACKAGING

5.1 Preservation - packaging.

5.1.1 Level C. Adapter & Fuze, Assembly: M624, (dwg. 12546032), shall be packaged as specified in 5.2.1.

5.2 Packing.

5.2.1 Level C. Twelve (12) Adapter & Fuze, Assembly: M624 (dwg. 12546032), shall be packed in accordance with dwgs. 12556001 and 12556005.

5.3 Marking. Marking shall be in accordance with dwg. 12556001.

5.4 Data cards. Data card information shall be as specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data. See MIL-A-48078.

6.2 Submission of inspection equipment designs for approval. (See MIL-A-48078). Equipment designs shall be submitted as required to: Commander, U S ARDEC, ATTN: AMSMC-QAT-I(D), Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the Contract. Unless otherwise specified, Data Item DI-R-1714 will apply.

6.3 Data cards. (See MIL-A-48078). Distribution of data cards shall include the following: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-M, Picatinny Arsenal, New Jersey 07806-5000.

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6.4 Drawings. Drawings listed in section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, US Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.6 Subject term (key word) listing.
Adapter
Fuze
Loaded Assembly
Energy output
Ammunition box
Wirebound box

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1345-A523)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-70608A (AR)	ADAPTER & FUZE, ASSEMBLY: M624, LOADING ASSEMBLING AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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